

OntoMPO: From Conceptual Model to MPO Ontology

Abstract

The Model for Project Observatories (MPO) provides a structured conceptual reference for the understanding and development of project observatory environments, organizing 61 concepts across three main dimensions: Structures, Processes, and Agents. However, the MPO is predominantly described in natural language, lacking a formal representation capable of rigorously specifying its concepts, relationships, constraints, and semantic assumptions, which creates room for interpretive ambiguities and inconsistencies in its application. This paper proposes OntoMPO, a domain ontology for the formal semantic representation of the MPO, developed following the Methontology methodology and grounded in the Unified Foundational Ontology (UFO) via OntoUML. The ontology was implemented in OWL using Protégé, resulting in 43 classes, 45 object properties, 59 data properties, 53 individuals, and 693 axioms. Evaluation was conducted through conceptual analysis and SPARQL-based competency question verification, confirming conceptual adherence to the MPO and logical consistency of the model. OntoMPO establishes a formally grounded and standardized conceptual vocabulary for the project observatory domain, contributing to the reduction of semantic ambiguities and providing a semantic infrastructure for future applications involving data integration, semantic querying, and comparative analysis across observatory initiatives.

Keywords

Ontologies, Project Observatories, Domain Ontology, MPO, OntoUML, UFO, Methontology

1. Introduction

Transparency and access to information have become increasingly relevant in the relationship between organizations and society, fostering accountability, trust, and more informed decision-making processes [1]. In project-intensive environments, transparency is particularly important, since projects constitute primary drivers of organizational transformation, innovation, and competitive advantage [2]. In this context, project observatories emerge as mechanisms for the centralization, analysis, and dissemination of project-related information, supporting evidence-based decision-making, organizational learning, and communication among stakeholders [3].

Among the initiatives proposed in this domain, the Model for Project Observatories (MPO), introduced by Vieira [4], provides a structured conceptual reference for the understanding and development of such environments. The MPO organizes 61 concepts distributed across three hierarchical levels – general, intermediate, and specific – systematically describing the structures, processes, and agents involved in project observatory initiatives.

Despite its conceptual robustness and empirical validation, the MPO is predominantly described in natural language, lacking a formal representation capable of rigorously specifying its concepts, relationships, constraints, and semantic assumptions. As a consequence, the model becomes highly dependent on human interpretation, creating room for semantic ambiguities, conceptual overlaps, and inconsistencies in its understanding and application. Different initiatives claiming adherence to the MPO may therefore adopt semantically incompatible interpretations while maintaining superficial alignment with the model's terminology, compromising its use as a stable conceptual reference for the analysis, comparison, and evolution of project observatories.

These limitations highlight the need for a more rigorous conceptual approach capable of explicitly representing the semantics underlying the MPO. In this context, ontologies have been widely adopted as mechanisms for formal knowledge representation, enabling the explicit specification of domain conceptualizations, semantic constraints, and formally grounded relationships [5, 6]. Formal ontological representations contribute to conceptual consistency, semantic interoperability, and the explicit repre-

sentation of ontological commitments across initiatives grounded in the same conceptual model [7]. However, although ontologies have been applied in different domains and observatory-related contexts, no studies were identified addressing the ontological formalization of project observatories or the semantic representation of conceptual models dedicated to this domain.

Therefore, this paper investigates the following research problem: *How can the Model for Project Observatories (MPO) be ontologically formalized in order to reduce semantic ambiguities and support a semantically rigorous and formally grounded representation of the domain?*

To address this problem, this paper proposes **OntoMPO**, a domain ontology for the formalization of the MPO. Developed following the Methontology methodology [8] and grounded in the Unified Foundational Ontology (UFO) [7] via OntoUML [9], the ontology systematically represents the concepts, relationships, attributes, and semantic assumptions underlying the model, making its ontological commitments explicit and providing a standardized and semantically well-defined conceptual vocabulary for project observatories. By formalizing the MPO, this work contributes to reducing conceptual ambiguities, improving interpretative consistency, and establishing a rigorous semantic foundation for the domain. The ontology is essentially conceptual in nature, focusing on domain formalization rather than on direct implementation in operational systems.

The remainder of this paper is organized as follows. Section 2 presents the theoretical background on project observatories, ontologies, and related work. Section 3 describes the research methodology. Section 4 presents the main results of the ontology construction. Section 5 concludes the paper and outlines directions for future work.

2. Theoretical Background

2.1. Project Observatories and the MPO

Project observatories are organizational mechanisms dedicated to the monitoring, analysis, and dissemination of project-related information, supporting transparency, organizational learning, and evidence-based decision-making. These environments systematically organize project data, metrics, indicators, and analytical resources in order to support the monitoring of project performance, processes, and results by different stakeholders [3]. In addition to supporting project management activities, project observatories also contribute to communication, collaboration, knowledge generation, and accountability practices within project-intensive environments [10].

Among the initiatives proposed in this domain, the Model for Project Observatories (MPO), introduced by Vieira [4], provides a structured conceptual reference for the understanding and development of project observatories. The MPO aims to systematize the main conceptual elements involved in project observatory environments by organizing 61 concepts distributed across three main dimensions: *Structures*, *Processes*, and *Agents*.

The *Structures* dimension encompasses elements associated with the structural organization of project observatories, including components, contents, and characteristics related to data organization and availability. The *Processes* dimension describes the procedures responsible for data collection, storage, sharing, and processing, providing a dynamic perspective of observatory operation. Finally, the *Agents* dimension represents the actors involved in observatory environments and the motivations influencing their interaction with projects and observatory resources.

By integrating structural, procedural, and social aspects, the MPO establishes a comprehensive conceptual foundation for project observatories. However, the model is predominantly described through natural language and conceptual descriptions, without a formal semantic representation capable of explicitly defining its concepts, constraints, and relationships. As a consequence, important conceptual distinctions and dependencies remain implicit, reinforcing the need for a semantically rigorous and ontologically grounded formalization of the domain.

2.2. Ontologies, UFO, and OntoUML

Ontologies have been widely adopted as mechanisms for formal knowledge representation and conceptual formalization, enabling the explicit specification of concepts, relationships, constraints, and semantic assumptions within a given domain [5, 6]. By establishing shared and semantically grounded conceptualizations, ontologies contribute to reducing ambiguities, improving conceptual consistency, and supporting interoperable representations of domain knowledge.

Among the foundational frameworks adopted for ontology-driven conceptual modeling, the Unified Foundational Ontology (UFO) [7, 9] provides a principled basis for the semantic analysis and representation of complex domains. UFO is organized into complementary sub-ontologies: UFO-A addresses endurants — persistent entities that maintain identity over time, including categories such as *Kind*, *Role*, *Phase*, and *Relator* — while UFO-C extends the framework to social entities, encompassing agents, goals, and normative descriptions. UFO distinguishes entities according to ontological properties such as identity, rigidity, dependence, and relationality, enabling more precise and semantically well-founded conceptual representations.

Based on UFO, OntoUML [11] extends traditional conceptual modeling languages by incorporating ontologically founded stereotypes and constraints. Unlike conventional UML models, OntoUML explicitly represents semantic distinctions through categories such as *Kind*, *Role*, and *Relator*. *Kinds* provide identity principles for their instances, whereas *Roles* represent context-dependent and relationally defined types. *Relators* mediate materially grounded relations among entities, enabling the representation of complex social and institutional relationships that cannot be adequately expressed by simple binary associations. These distinctions contribute to reducing semantic ambiguities and improving conceptual consistency in domain representations.

The adoption of UFO and OntoUML is particularly relevant in domains characterized by complex organizational, social, and informational relationships, such as project observatories. In this context, ontological grounding supports the explicit representation of conceptual dependencies, actor participation, informational structures, and interaction processes underlying the MPO domain.

2.3. Ontology Engineering: Methontology

Developing a domain ontology requires a principled methodology that guides the construction process from initial scoping through evaluation. Methontology [8], proposed by Fernández-López, Gómez-Pérez, and Juristo, is one of the most widely recognized methodologies for ontology engineering, providing a systematic framework that covers the entire ontology lifecycle. The methodology defines management and support activities that accompany all stages of development, alongside six main construction stages: specification, conceptualization, formalization, integration, implementation, and evaluation.

Each stage produces well-defined artifacts. Specification establishes the ontology's purpose, scope, and competency questions. Conceptualization organizes domain knowledge into language-independent structures such as term glossaries, concept classification trees, verb diagrams, and rule tables. Formalization translates these structures into a formal representation. Integration promotes reuse of existing ontologies and models. Implementation encodes the formalized ontology in a computational language. Evaluation verifies the artifact's quality against the initial requirements.

The choice of Methontology for this work is justified by its systematic and iterative character, its explicit support for the production of intermediate conceptual artifacts that facilitate domain analysis and validation, and its compatibility with OntoUML-based formalization workflows. Although Methontology[8] was proposed in 1997, it remains widely adopted in ontology engineering due to the clarity of its lifecycle stages and the language-independent nature of its conceptual artifacts — properties particularly relevant when formalizing a conceptual model described predominantly in natural language, as is the case of the MPO. More recent alternatives were considered but not adopted: modular-ontology methodologies such as NeOn are better suited for networked ontology ecosystems, while process-oriented approaches lack the structured conceptualization artifacts required to systematize the MPO domain prior to formalization. In this context, Methontology provided the most adequate

balance between methodological rigor, conceptual expressiveness, and practical applicability for the objectives of this work.

2.4. Related Work

Although ontologies have been widely employed for semantic representation and knowledge organization in different domains, few studies have explored their application in observatory-related environments. Existing works primarily focus on interoperability, semantic integration, and knowledge representation in scientific and analytical contexts.

Chen and Chen [12] proposed an ontology-based multi-agent framework to support information retrieval and processing in auroral research observatories, enabling semantic standardization and matchmaking between user requests and available services. Fox et al. [13] developed the Virtual Solar-Terrestrial Observatory (VSTO), employing ontologies and Semantic Web technologies to support semantic interoperability and integrated access to distributed scientific data in solar physics. In the environmental domain, Masmoudi et al. [14] proposed a modular ontology grounded in the Basic Formal Ontology (BFO) to support semantic interoperability in environmental monitoring systems. More recently, Yoshiura et al. [15] introduced a conceptual model for Health Observatories based on Semantic Web technologies, emphasizing semantic organization and evidence-based decision support.

Despite their contributions, these studies focus on scientific, environmental, and health-related observatory contexts, without addressing the specific conceptual structures, processes, and organizational agents involved in project observatories. Furthermore, none of the identified studies proposes an ontological formalization grounded in the MPO or supported by foundational ontologies such as UFO and OntoUML. Table 1 summarizes the main characteristics of the identified works and contrasts them with the proposal presented in this paper.

Table 1
Comparison between related works and the proposed approach.

Work	Domain	Ontology Usage	Observatory Context	Project Domain
Chen and Chen [12]	Auroral research	Semantic representation	Scientific	No
Fox et al. [13]	Solar physics	Semantic interoperability	Scientific	No
Masmoudi et al. [14]	Environmental monitoring	BFO-based ontology	Environmental	No
Yoshiura et al. [15]	Health observatories	Semantic organization	Health	No
This work	Project observatories	UFO/OntoUML	Project-based	Yes

The analysis of related works reveals an important research gap regarding the ontological formalization of project observatories. In particular, no studies were identified addressing the semantic formalization of the MPO or proposing a foundationally grounded conceptual representation for this domain — one supported by UFO and modeled in OntoUML. This work contributes to filling this gap by proposing a domain ontology that explicitly represents the concepts, relationships, and semantic constraints underlying the MPO.

3. Methodology

This research aims to construct a domain ontology for the formalization of the MPO [4]. From a research classification standpoint, the work is of a qualitative nature, grounded in an interpretivist approach,

given its focus on the analysis, comprehension, and explicit representation of an existing conceptual model. The investigation relies primarily on documentary research, drawing on the official MPO documents, correlated scientific publications, and specialized literature on ontologies and observatories. Data analysis was conducted through content analysis, with the purpose of identifying, organizing, and systematizing the concepts, relations, and attributes relevant to the domain.

Regarding its ontological classification, OntoMPO is deliberately designed as a *lightweight ontology* [6]. Lightweight ontologies are characterized by the use of hierarchical taxonomies, binary relations, and basic property constraints, without resorting to highly expressive logical axiomatizations such as complex role chains, nominals, or concrete domains. This design choice was intentional and motivated by three complementary factors. First, the primary objective of OntoMPO is the formal *representation* of the MPO conceptual domain — making its concepts, relations, and semantic assumptions explicit and unambiguous — rather than supporting deep automated inference or theorem proving. Second, a lightweight design favors ontology accessibility and reusability: a leaner formal structure is more readily understandable by domain practitioners, extensible by future specialized ontologies, and integrable with external systems without imposing high computational overhead. Third, the MPO itself is a conceptual reference model, and its formalization at this stage is intended to establish a stable and shared semantic foundation for the domain, on top of which heavier axiomatizations can be incrementally added in future work as specific application requirements demand. In terms of generality, OntoMPO operates as an *intermediate ontology*, bridging foundational and application-level representations. As to its purpose, it is classified as a *domain ontology*, as it formalizes knowledge specific to the context of project observatories.

The research was structured in two main stages. In the first, an exploratory literature review was carried out to provide theoretical grounding on observatories, transparency, ontologies, and ontological engineering methodologies. In the second stage, the ontology was developed following the guidelines of the Methontology methodology [8], as described in the subsections below.

3.1. Methontology

Methontology defines a systematic set of activities for ontology construction, encompassing management and support activities that accompany the entire lifecycle of the ontological artifact. Alongside these, the methodology establishes six main development stages — specification, conceptualization, formalization, integration, implementation, and evaluation — organized in a continuous cycle of model evolution [8].

3.1.1. Specification

The specification stage aims to define the scope and requirements of the ontology, identifying the domain of interest, the objectives of the ontological model, and the main stakeholders involved [8]. In this work, the ontology was specified to formally represent the central concepts of the MPO, making the domain knowledge computationally interpretable while preserving the semantics of the original conceptual model and reducing the ambiguities inherent to its natural-language description.

The ontology covers the three MPO layers — *Structures*, *Processes*, and *Agents* — including their general, intermediate, and specific concepts. Object properties capture cross-layer relations (e.g., *Agent executes Process*, *Process produces Content*, *Structure supports Process*), while data properties describe literal attributes. The primary knowledge source is the MPO conceptual model itself [4].

The ontology's competency questions, which define the minimum set of queries the artifact must be capable of answering, were formulated as follows: (i) How are the concepts hierarchically organized within each MPO dimension? (ii) How is the Structures layer represented? (iii) How is the Processes layer represented? (iv) How is the Agents layer represented? (v) What relations exist among structures, processes, and agents?

3.1.2. Conceptualization

The conceptualization stage structures domain knowledge independently of any implementation language, organizing the concepts, relations, and attributes identified during specification. In this work, the following Methontology artifacts were produced: (a) a *term glossary*, gathering all relevant domain resources and their meanings; (b) *concept classification trees*, organizing domain concepts into taxonomies for each MPO layer; (c) *verb diagrams*, expressing the declarative meaning of relations; and (d) *formula and rule tables*, formalizing constraints, axioms, and derivable rules. These artifacts provided a shared, language-independent representation of the domain that served as the foundation for the subsequent formalization stage.

3.1.3. Formalization and Integration

Although Methontology treats formalization and integration as separate stages, they were conducted in an iterative and partially parallel manner in this work. Integration decisions — particularly the grounding of concepts in UFO categories — directly shaped formalization choices, and conversely, the formalization process in OntoUML prompted refinements in the integrated concepts. This incremental approach is consistent with Methontology’s iterative character and contributed to a more semantically coherent ontological model.

The **formalization** stage transforms the conceptual model into a formal, unambiguous representation [8]. OntoUML [7] was adopted as the formalization language, given its ability to rigorously represent conceptual elements through ontologically founded stereotypes derived from UFO. Concepts from the Structures, Processes, and Agents dimensions were modeled as ontological classes following UFO categories such as *Kind*, *Role*, *Relator*, and *Mode*. Relations (e.g., *executes*, *generates*, *supports*) were formalized as ontological associations with explicit domains, ranges, multiplicities, and restrictions. Attributes were described as intrinsic or relational properties aligned with the ontological categories of their associated classes.

The **integration** stage promotes systematic reuse of ontologies and conceptual models [8]. Integration with the MPO occurred primarily during specification and conceptualization, where domain concepts were identified and structured from the reference model. In the integration stage proper, the focus shifted to ontological grounding via UFO, enabling the correct classification of concepts according to its categories and ensuring semantic coherence throughout the model. This integration directly supported the OntoUML formalization, as modeling decisions were guided by well-established ontological principles, reducing ambiguities and favoring the ontology’s consistency and extensibility.

3.1.4. Implementation

Consistent with the lightweight ontology design adopted for OntoMPO, the implementation prioritizes semantic clarity and structural coverage over logical expressiveness. Rather than defining complex axioms or inference rules, the OWL encoding focuses on explicit class hierarchies, object and data property definitions, domain and range restrictions, and individual assertions — elements sufficient to support semantic querying and consistency checking while keeping the model computationally tractable and accessible for future extension.

The implementation stage materializes the formalized ontology in a computational environment, translating the ontological model into an executable representation [8]. In this work, the ontology was implemented using Protégé, a widely adopted tool for ontology development and management. The OntoUML model, grounded in UFO, was translated into OWL 2 DL, a language that supports explicit definition of classes, properties, restrictions, and axioms, as well as automated consistency checking via description-logic reasoners. Ontological classes were encoded as OWL classes, relations as object properties, and attributes as data properties, with their respective domains, ranges, and restrictions.

3.1.5. Evaluation

The evaluation stage verifies the quality of the developed ontology, ensuring that it meets the requirements defined during specification and is semantically coherent and consistent with the domain [8]. In this work, evaluation was conducted along two complementary dimensions.

The **conceptual and ontological evaluation** consisted of a manual inspection of the model based on OntoUML modeling guidelines and UFO ontological categories [7]. This analysis verified the correct classification of concepts, the coherence of taxonomic hierarchies, the adequacy of cross-dimension relations, and the absence of ambiguities or conceptual contradictions, thereby assuring semantic fidelity to the MPO domain.

The **functional evaluation** was conducted using a structured set of competency questions derived from the MPO dimensions, covering: (i) the conceptual structure of the observatory; (ii) relations among Structures, Processes, Agents, and Contents; (iii) roles performed by agents; (iv) processes and their information flows; and (v) the structural and informational elements involved. To verify whether the ontology is capable of answering these questions, SPARQL queries were executed over the OWL implementation, and the returned results were analyzed against the expected answers according to the MPO domain. This procedure allowed the assessment of the ontology's capacity to support semantic queries and fulfill the informational requirements established during specification.

4. Results

This section presents the main results obtained across the Methontology development stages, covering conceptualization, formalization and integration, implementation, and evaluation.

4.1. Conceptualization

The conceptualization stage produced a set of language-independent artifacts systematizing the knowledge of the MPO domain across the Structures, Processes, and Agents dimensions. The artifacts produced include: a *term glossary* with **53 terms**; *concept classification trees* supported by a data dictionary with **17 concepts**, **41 instance attributes**, **40 class attributes**, **20 constant tables**, and **53 instance tables**; a *verb diagram* with a dictionary of **15 verbs** and **67 condition entries**; and *formula tables* with **55 formulas**.

4.2. Formalization and Integration

The formalization of OntoMPO was carried out using OntoUML [11], grounded in UFO [7]. Integration with UFO occurred intrinsically to the formalization process: using OntoUML implies automatic alignment of modeled concepts with UFO ontological categories, so that OntoMPO inherits widely accepted ontological principles without requiring explicit mappings to external ontologies. The MPO's three layers were modeled through complementary diagrams at different levels of granularity, including cross-layer and intra-layer views.

Modeling decisions reflect deliberate ontological design choices grounded in UFO categories. Concepts whose existence is constituted by the mediation between two or more participants were systematically modeled as *Relators* [7]. *Observation* is a representative case: it cannot exist without an Agent that observes and a Knowledge entity as its epistemic result, making its meaning ontologically dependent on both participants. Similarly, *Connection* was modeled as a Relator because, in the MPO, a network connection carries attributes — such as type, topology, and scope — that belong neither to Hardware nor to Network in isolation, but emerge only from their mediation; the same reasoning applies to *Operation*. *SocialInteraction*, *CrudOperation*, and *Log* follow the same rationale, as each requires the co-participation of an Agent and a structural counterpart to be semantically grounded.

In the Processes layer, *Extract*, *Transform*, and *Load* were also modeled as Relators rather than events. Although UFO-B provides constructs for modeling events with temporal participation, adopting

it for ETL stages would introduce complexity – including temporal participation and event mereology – that exceeds the granularity and scope of OntoMPO as a lightweight ontology. In the MPO, each ETL step is defined essentially by the relation it mediates between roles: Extract connects a DataSource and a Collector, Transform connects a Collector and a Processor, and Load connects a Processor and a Storer. The Relator captures this relational dependency without requiring full event modeling. Finally, ProjectObservatory was classified as a *SubKind* of DataManager because it inherits the same identity principle – both manage project data – without introducing a new one; and ObservatoryGroup was modeled as a *Collective* because its existence presupposes member Agent instances bound by «MemberOf» relations, reflecting a genuinely collective rather than merely aggregate ontological structure.

Figure 1 presents the complete OntoUML diagram, integrating all three MPO layers and demonstrating how Structures, Processes, and Agents articulate to represent the observatory domain as a whole.

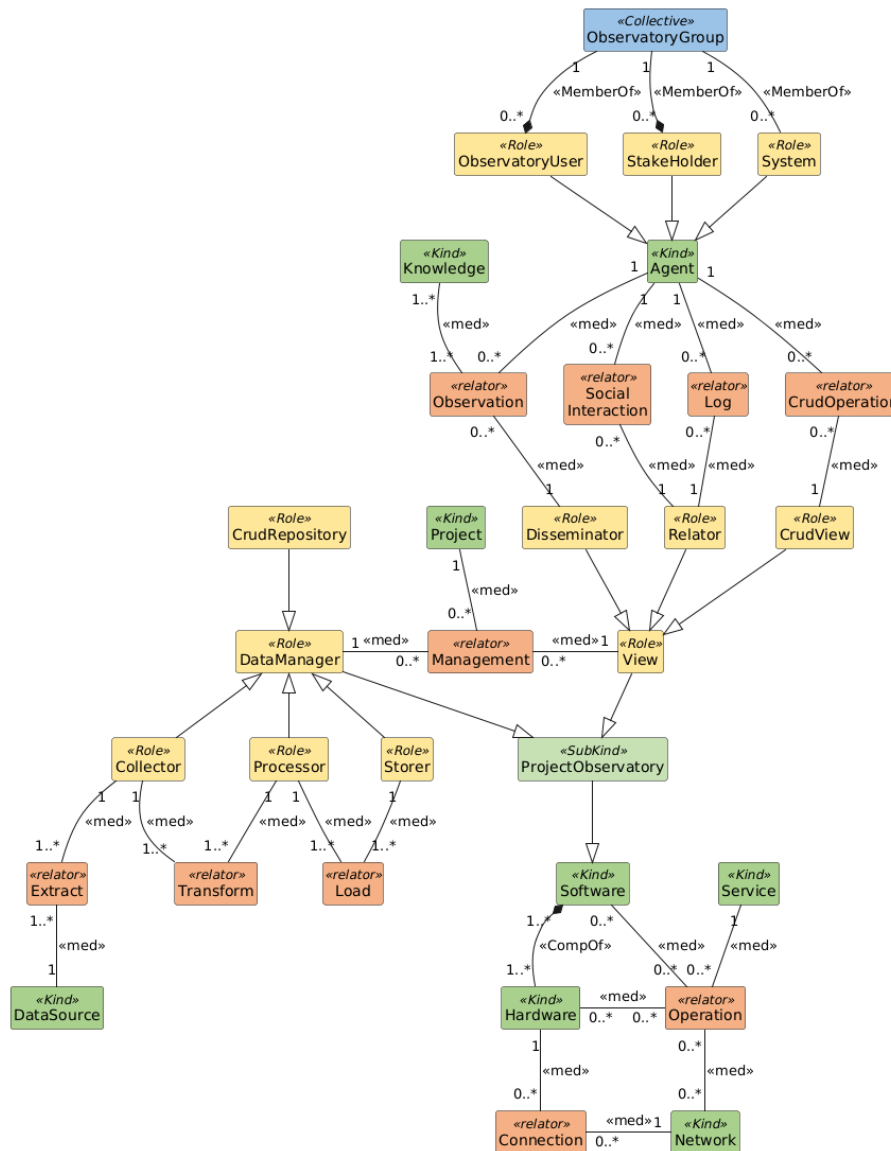


Figure 1: Complete formalization of the MPO in OntoUML. Source: Author's own work.

4.3. Implementation

The OntoUML model was implemented in OWL using Protégé. This step enabled syntactic and semantic validation of the model, explicit axiom definition, and the expression of formal constraints not fully representable in conceptual models. The resulting ontology comprises **43 classes**, **45 object properties**, **59 data properties**, and **53 individuals**. In terms of logical expressiveness, the model contains a total of **693 axioms**, of which 434 are logical axioms and 198 are declaration axioms, reflecting a significant degree of semantic formalization that extends beyond a purely taxonomic structure. These metrics indicate that the ontology incorporates restrictions, relations, and formal definitions that enable automated reasoning while maintaining a balance between expressiveness and computational complexity. Figure 2 presents the OntoMPO developed using Protégé.

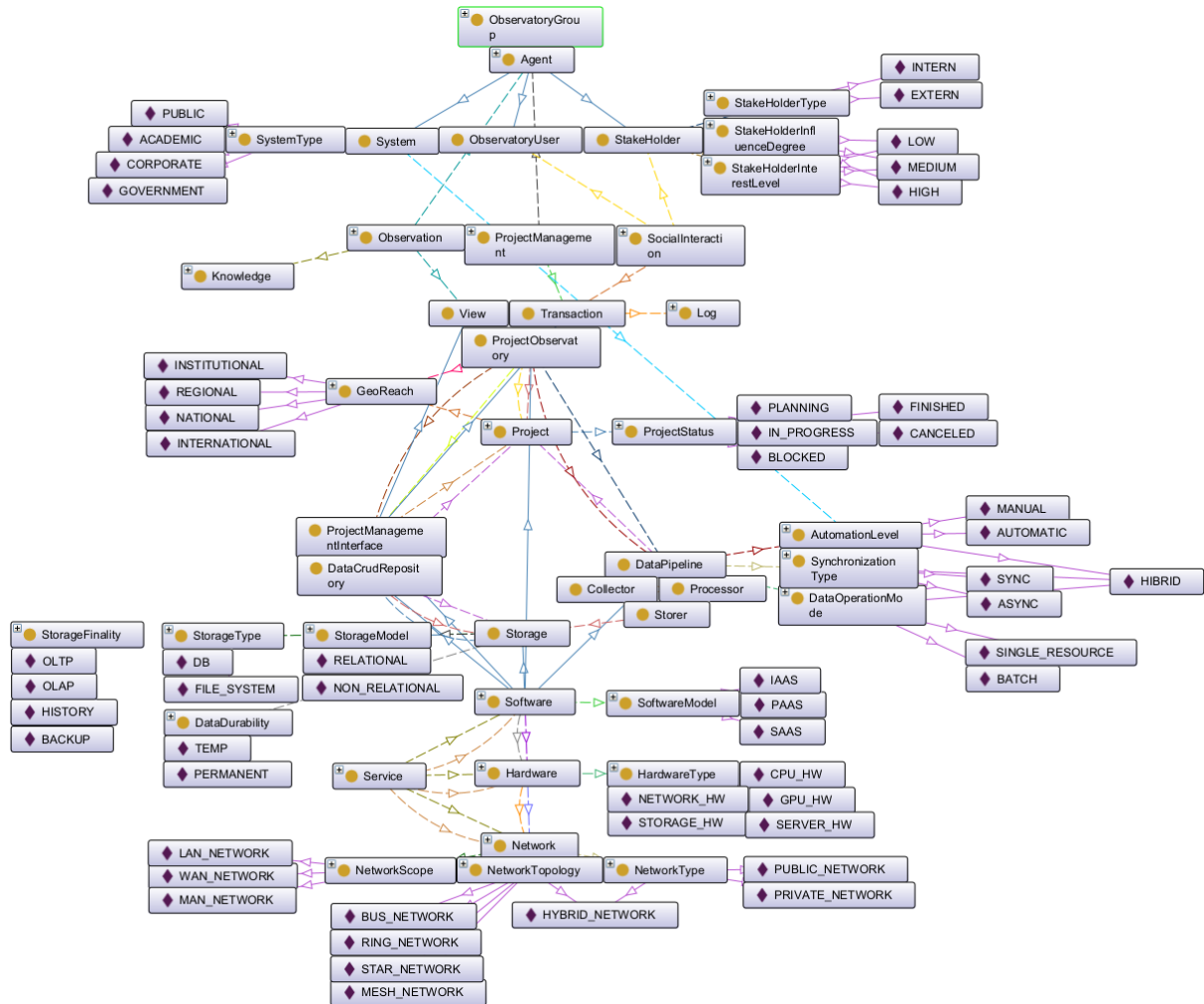


Figure 2: OntoMPO developed using Protégé. Source: Author's own work.

4.4. Evaluation

The evaluation of OntoMPO was structured along two complementary dimensions: conceptual evaluation and competency-question-based evaluation.

4.4.1. Conceptual Evaluation

The conceptual evaluation verified the ontology's coverage, semantic coherence, and correspondence between the MPO reference model and the formalized elements. The main concepts of the MPO –

agents, processes, and structures — are explicitly represented in the ontology, preserving the semantics of the original conceptual model. The definition of ontological properties and restrictions made semantic relations explicit, such as agent participation in processes and the association between organizational structures and the contents they produce. Overall, the ontology presents conceptual adherence to the MPO, offering a consistent and semantically clear formal representation of the domain.

4.4.2. Competency-Question-Based Evaluation

The competency-question evaluation verified whether the ontology can answer the five representative domain queries defined during specification. SPARQL queries were executed over the OWL implementation and results were post-processed using a Python script that organizes classes into hierarchical structures and identifies semantic relations based on ontological properties.

QC1 was addressed by retrieving specialization relations via `rdfs:subClassOf`, confirming a well-defined hierarchical structure in which general concepts are progressively specialized. QC2, QC3, and QC4 were addressed by rooting the same query at `Software`, `ProjectManagement`, and `Agent`, respectively, revealing coherent hierarchical organizations for the Structures, Processes, and Agents layers. For QC5, object properties were extracted via `owl:ObjectProperty` with their domains and ranges, showing that cross-layer connections are frequently mediated by central concepts such as `Observation` and `Knowledge`. Figure 3 illustrates an excerpt of this cross-layer relation analysis, demonstrating that the MPO layers are connected not only by direct relations but also through semantic mediation mechanisms that ensure cohesion and modularity in the ontological structure.

```
Observation
[AS DOMAIN]
  → observation_is_available_by → View
  → observed_by → Agent
  → produces_knowledge → Knowledge

Knowledge
[AS RANGE]
  ← produces_knowledge ← Observation
```

Figure 3: Excerpt of the cross-layer relation analysis result. Source: Author’s own work.

4.4.3. OntoMPO Summary

As the main result of this research, OntoMPO is a domain ontology that formally represents the concepts, relations, constraints, and semantic assumptions of the MPO. Developed following Methontology [8], formalized in OntoUML [9] grounded in UFO [7], and implemented in OWL, OntoMPO provides a formal conceptual foundation that reduces the ambiguities present in the original natural-language description of the MPO and establishes a structured vocabulary for the project observatory domain. This formalization contributes to conceptual standardization and creates conditions for future applications, such as observatory management support systems, knowledge-sharing platforms, and comparative analysis mechanisms across different observatory initiatives.

5. Conclusion

OntoMPO is a domain ontology that formally represents the conceptual structure of the Model for Project Observatories (MPO) [4], making explicit the semantic relations, ontological commitments, and conceptual constraints that remain implicit in its original natural-language description. By providing a formally grounded and computationally interpretable representation of the MPO, OntoMPO establishes

a shared semantic reference for the project observatory domain, contributing to conceptual standardization, reduction of interpretive ambiguities, and improved consistency across initiatives that claim adherence to the model. These benefits are particularly relevant in project-intensive organizational environments, where the lack of a formal semantic foundation has historically allowed structurally divergent observatory implementations to coexist under the same conceptual umbrella without a rigorous basis for comparison or alignment.

The development of OntoMPO produced a comprehensive set of conceptual and formal artifacts that collectively constitute a structured knowledge base for the project observatory domain. During the conceptualization stage, the following artifacts were generated: a *term glossary* with 53 terms providing precise definitions for all domain concepts; a *data dictionary* with 17 concepts, supported by *instance* and *class attribute tables* with 41 and 40 entries respectively, and *constant tables* with 20 entries; a *concept classification tree* organizing domain concepts into taxonomic hierarchies across the Structures, Processes, and Agents dimensions; a *verb dictionary* with 15 verbs and an associated *condition table* with 67 entries formalizing the semantics of domain relations; and *formula tables* with 55 entries capturing derivable rules and constraints. In the formalization and implementation stages, these artifacts were elevated into a full ontological model: an *OntoUML conceptual model* grounded in UFO [?], representing domain concepts through ontologically founded stereotypes such as *Kind*, *Role*, *Relator*, and *Collective*; and an *OWL implementation* in Protégé comprising 43 classes, 45 object properties, 59 data properties, 53 individuals, and 693 axioms, of which 434 are logical axioms.

The main contributions of this work are threefold. First, the ontological formalization of the MPO as a semantically rigorous artifact, transforming a conceptual model described in natural language into a formally grounded representation with explicit classes, properties, restrictions, and axioms. Second, the provision of a standardized conceptual vocabulary for the project observatory domain that can serve as a shared reference for researchers, practitioners, and system developers working with MPO-based initiatives. Third, the establishment of OntoMPO as a semantic infrastructure for the domain, enabling future applications involving semantic querying, data integration, automated classification, and comparative analysis across observatory implementations.

This work presents limitations that should be acknowledged. The evaluation of OntoMPO was conducted through conceptual analysis and SPARQL-based competency question verification, without structured validation sessions involving domain experts, which would further strengthen confidence in the ontology's coverage and semantic adequacy. Additionally, given the conceptual nature of OntoMPO — focused on domain formalization rather than operational deployment — automated inference capabilities were not explored in depth. The ontology was designed to formally represent the MPO conceptual structure, and its current expressiveness in OWL 2 DL, while sufficient for semantic querying and consistency checking, was not extended to support complex rule-based reasoning or role chain inference. These aspects delimit the scope of the present work without compromising its primary objectives.

Future work envisions the progressive construction of an *ontology ecosystem for the MPO*, in which OntoMPO serves as a foundational layer to be extended by specialized ontologies addressing specific observatory application contexts, organizational settings, and project domains. This ecosystem would include alignments with related ontologies from the project management domain, integration with real-world observatory datasets, and the definition of more expressive axioms and inference rules to support deeper semantic reasoning. A particularly relevant direction concerns the role of ontologies as a semantic foundation for artificial intelligence applications. The formally grounded vocabulary and explicit semantic structure of OntoMPO can serve as a knowledge layer to support and constrain generative AI models in project observatory contexts, improving domain consistency and reducing the risk of semantically incoherent outputs. Conversely, large language models can be explored as instruments for ontology population, automatic concept extraction from unstructured observatory documents, and natural language interfaces for querying the OWL implementation. This convergence between formal ontological knowledge and generative AI represents a frontier with significant potential for advancing both the theoretical foundations and the practical applicability of the MPO ecosystem in project-intensive organizational environments.

6. Appendices

- Github repository with the artifacts generated in this research;

Declaration on Generative AI

During the preparation of this work, the author(s) used Claude (Anthropic) in order to: writing assistance, grammar and spelling check, and academic English revision. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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